(iii) Place value grid model

So far we have discussed grid models to do addition and subtraction of decimal numbers. Earlier we have studied representation of decimal numbers in place value tables. Let us use the same representation for addition and subtraction of decimal numbers.

For example, while adding 4.83 and 1.67, we have

Decimal No.	Ones	Tenth	Hundredth
4.38	4	3	8
1.67	1	6	7
6.05	6	0	5

Therefore, 4.38 + 1.67 = 6.05.

Example 1.10 Add the following: (i) 30.9 + 52.73 (ii) 25.67 + 33.856

Solution

(i) 30.9 + 52.73

Let us use the place value grid.

Decimal No.	Tens	Ones	Tenth	Hundredth
30.9	3	0	9	0
52.73	5	2	7	3
83.63	8	3	6	3

(Since the digits in the decimal place of 52.73 is 2 and 30.9 is 1, we should add 0 at the hundredth place of 30.9 to equalise the digits in the decimal place)

Therefore, 30.9 + 52.73 = 83.63.

(ii) 25.67 + 33.856

Let us use the place value grid.

Decimal No.	Tens	Ones	Tenth	Hundredth	Thousandth
25.67	2	5	6	7	0
33.856	3	3	8	5	6
59.526	5	9	5	2	6

Therefore, 25.67 + 33.856 = 59.526.

Note

Adding zeros at the right end of decimal digits will not change the value of the number.

Example 1.11 Everyday Malar travels 1.820 *km* by bus and 295 *m* by walk to reach the school. Find the distance of school from her house in *km*.

Solution

Distance travelled by bus =1.820 *km*

Distance covered by walk = 0.295 *km*

Total distance = 1.820 + 0.295

=1.820 + 0.295

=2.115 *km*

1000 *m*=1 *km*; 1 *m* = $\frac{1}{1000}$ *km*

Hence, 295 *m* = $\frac{295}{1000}$ *km*

= 0.295 *km*

Therefore, the school is situated at the distance of 2.115 *km* from her house.

7th_Maths_T3_EM_NS-Chp1.indd 8 05-11-2019 17:44:50